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Review

Managing salinity and waterlogging in the Indus Basin of Pakistan

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ABSTRACT

Waterlogging and salinization are major impediment to the sustainability of irrigated lands and livelihoods of the farmers, especially the smallholders, in the affected areas of the Indus Basin. These problems are the result of a multitude of factors, including seepage from unlined earthen canals system, inadequate provision of surface and subsurface drainage, poor water management practices, insufficient water supplies and use of poor quality groundwater for irrigation. About 6.3 million ha are affected by different levels and types of salinity, out of which nearly half are under irrigated agriculture. Since the early 1960s, several efforts have been made to improve the management of salt-affected and waterlogged soils. These include lowering groundwater levels through deep tubewells, leaching of salts by excess irrigation, application of chemical amendments (e.g. gypsum, acids, organic matter), and the use of biological and physical methods. However, in spite of huge investments, the results have in general been disappointing and the problems of waterlogging and salinity persist.

This paper reviews sources, causes and extent of salinity and waterlogging problems in the Indus Basin. Measures taken to overcome these problems over the last four decades are also discussed. The results reveal that the installed drainage systems were initially successful in lowering groundwater table and reducing salinity in affected areas. However, poor operation and maintenance of these systems and provision of inadequate facilities for the disposal of saline drainage effluent resulted in limited overall success. The paper suggests that to ensure the sustainability of irrigated agriculture in the Indus Basin, technical and financial support is needed and enhanced institutional arrangements including coordination among different federal and provincial government agencies to resolve inter-provincial water allocation and water related issues is required.

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1. Introduction

Irrigated agriculture in Pakistan is mainly confined to the Indus plains where it has been developed by harnessing the principal water resources available to the country. The contiguous Indus basin irrigates an area of about 16 million ha. Under the Indus water treaty of 1960 the total water available to Pakistan from the Indus River system was 181×10^9 m^3, approximately 75% of the annual available flow (World Bank, 2005). Of the available water, about 131×10^9 m^3 is diverted to the 43 main canal irrigation systems while 11×10^9 m^3 is lost in the river system and 39×10^9 m^3 flow to the sea (Badruddin, 1996). The total length of the canals, main canal branches, and distributaries is about 57,000 km. The system has 88,600 outlets for the irrigation of service areas. The length of the farm channels and watercourses is about 1.6 million km (Bhatti and Kijne, 1992).

Due to arid and semi-arid conditions prevailing in most of the Indus Basin, the evapotranspiration demand is high and rainfall is scanty and sporadic, limiting productive agricultural systems within the basin to those areas which can be assured of a reliable water supply. Water use efficiencies within the irrigated areas are generally low, with much of the diverted surface water recharging the underlying aquifers. These losses in the upstream of the system have serious consequences for the delivery of surface water to the farmers located at the end of the canal system. However, at the upstream of the system, this groundwater has increasingly been exploited to supplement the surface water supplies, resulting in an increasingly higher system and basin level efficiency (Shah, 2007). However it should be noted at this point that irrigated agriculture productivity remains relatively low (Ahmad et al., 2007).

Over the last decade, the number of private tubewells in Pakistan has taken a quantum jump mainly because of decreasing surface water supplies and incidences of drought. During the extreme drought conditions during 1996–2001, the surface water availability in Pakistani Punjab was reduced by 46%. This led to a 59% increase in the number of private tubewells over the same period (Qureshi et al., 2003). Most of these tubewells were funded by farmers except few community tubewells which were installed by NGOs to provide water for domestic purposes. However, government provided relief to local farmers by exempting surface water fee for the drought period. Due to over-pumping during this period, groundwater table dropped in many areas especially in Baluchistan province which was worst hit by drought. The decline of groundwater table in Baluchistan was up to 2–3 m per year (Qureshi et al., 2004). As a result, groundwater became inaccessible to small poor farmers as they were not able to deepen their pumps and install large capacity power supplies. According to government estimates, the water table dropped to depths that were inaccessible in 5 and 15% of the irrigated areas in the Punjab and Baluchistan provinces, respectively, during this period (PPSGDP, 2000).

Despite improvements in surface water supplies after 2001, installation of private tubewells continued and currently about 0.8 million small capacity private tubewells are working in Pakistan. It is estimated that investment on the private tubewells is of the order of Rs. 25 billion (US $ 400 million) whereas the annual benefits in the form of agricultural production are to the tune of Rs. 150 billion (US $ 2.5 billion) (Shah et al., 2003). The estimated number of users is over 2.5 million farmers, who exploit groundwater directly or hire the services of tubewells from their neighbors. The groundwater is currently providing more than 50% of the total crop water requirements with the flexibility of its availability as and when needed (Shah, 2007).

These tubewells are used for supplying supplemental flows for irrigation and to provide much of the drinking water for urban areas. The total groundwater abstraction from these tubewells is estimated at 45×10^9 m^3 against a recharge of 40–60×10^9 m^3 (Shah et al., 2003). Out of this, about 33×10^9 m^3 is extracted through private small capacity tubewells, which are mostly located in fresh groundwater areas to capture fresh seepage from irrigation canals. The remaining 12×10^9 m^3 is extracted by large capacity public tubewells mainly to provide domestic water supplies to urban areas.

The small capacity shallow tubewells (up to 6 m depth) were initially installed by farmers to capture the seepage from unlined canals to supplement irrigation supplies for meeting crop water requirements. Therefore their installation and operational costs were very low and farmers were enjoying their benefits without much financial burden. With the increasing groundwater table depth (>15 m), farmers have to switch from small diesel operated tubewells to high powered electric and diesel engines. Most of the electric tubewells were installed in 1970s and 1980s when government was providing subsidies on the installation costs. Due to increasing energy prices, government withdrew subsidies in the early 1990s, which resulted in stagnation of the growth of electric tubewells and increase in diesel engine operated tubewells. According to recent estimates, only 13% of the total

tubewells are operated by electric motors whereas the rest 85% are operated by diesel engines of different capacities[2] (Qureshi and Akhtar, 2003). Diesel pumps are preferred by farmers because of their low installation costs as compare to electric tubewells. Other advantages of diesel tubewells include timely availability of water (no power cuts), suitability for farmers having small holdings and fragmented land (due to moveable diesel engine) and non requirement of any reserved money (no tension of monthly bill).

The on-demand availability of fresh groundwater has helped the farmers to cope with the vagaries of the surface water supplies and achieve secure and predictable yields. However, one undesirable consequence of intensive use of groundwater has been the development of secondary salinization. As a result, large tracts of irrigated lands are already salinized and many others are under threat. It is estimated that nearly 4.5 million hectares are already afflicted with this menace, of which about half are located in irrigated areas (WAPDA, 2003). About 30% of this area lies in the Punjab province, which produces more than 90% of the country's total food production. Another 1.0 million hectare is affected by water logging. Of the salt-affected soils in Pakistan, about 40% are saline (containing elevated levels of soluble salts) while 60% are saline-sodic and sodic in nature, which are characterized by elevated levels of sodium (Na^+). These serious environmental problems have become a great challenge to ensure food security for the ever increasing population of Pakistan.

Since the early 1960s, numerous efforts have been made to cope with the problems of waterlogging and salinity. These included lining of watercourses to control seepage losses, adaptation of improved irrigation practices and the installation of surface and subsurface drainage systems. Most of these projects were funded by government and no investments were made by farmers. Over the last three decades, considerable work has also been done on the reclamation of saline soils through physical, chemical and biological strategies. Due to its relatively low price, general availability and easy application, gypsum ($CaSO_4{\cdot}2H_2O$) is most commonly used for the reclamation of saline-sodic and sodic soils. During 1970s, government introduced subsidy on gypsum to encourage farmers for the rehabilitation of affected soils. This scheme proved very successful however, these efforts could not be converted into a national program. These efforts remain confined to support on-farm research on salinity and sodicity management. On the other hand, farmers continue their efforts to reclaim salt affected lands through improved water management, crop choices and cultural practices. However these efforts were limited to field scale level and have generally not been taken up by the wider farming community.

Despite huge investment, the success has been limited in solving land degradation issues. The problems of waterlogging and salinity could not be solved as initially envisaged. About 15–20% of the land still suffers from salinity and another 20–30% is confronted with high groundwater levels, from the over application of surface water. Soil salinity problems are particularly serious in the Sind province where some 70–80% of the soil is classified as moderately or severely saline (Smedema, 2000).

This paper discusses the causes and extent of salt-affected and waterlogged soils in the Indus Basin of Pakistan. It also reviews the public and private sector interventions made in the past for the management of saline and waterlogged soils in the Pakistan. The lessons learned and future perspectives of managing salt-affected and waterlogged soils are also the foci of this paper.

2. Irrigation development in the Indus Basin

About a century ago, Indus Basin Irrigation System (IBIS) was originally designed for an annual cropping intensity (i.e. cropped area per year) of 75% with the intention of spreading water to as large an area as possible to expand the settlement opportunities. The major objective of irrigation development at that time was to prevent crop failure and avoid famine (Jurriens and Mollinga, 1996). Another design feature of IBIS was low management and operational requirements, which is an advantage with an inherited disadvantage of inflexibility.

The IBIS typically consists of the main canals from which the water is distributed to branch canals. Secondary canals, called distributaries, take off from the branch canals. The distributaries and their branches, called minors are the main arteries for releasing water to small irrigation areas (averaging 160 ha). The outlets are free draining structures, which have a capacity fixed in proportion to the service area. Due to variations in the main canal discharges, it becomes difficult to achieve equity in water distribution. In times of water shortage, water has to be rotated between secondary canals, the distributaries and minors.

The operation of the Indus basin irrigation system is based on a continuous water supply and is not related to actual crop water requirements. Irrigation canals are usually not allocated more than their design capacity, of which a typical value is about $2\ mm\ d^{-1}$. Despite significant increase in storage capacities, it is essentially a supply-based system. Hence, it cannot adequately accommodate changing water demands during the crop season.

Water distribution to farmers is based on a seven day fixed rotational turn called ''warabandi''. This means that each farmer is allowed to take an entire flow of the outlet once in seven days and for a period proportional to the size of his land holding. The amount of water during a turn is usually insufficient to irrigate the entire farm. Farmers usually practice either deficit irrigation to irrigate their entire land or choose to leave a fraction of their land holding un-irrigated. Due to age and poor maintenance, the delivery efficiency of irrigation system is low, ranging from 35 to 40% from canal head to the crop root zone (Tarar, 1995). In practical terms, therefore, much of the surface water is currently lost en-route, which has serious implications for downstream farmers located at the end of the canal system.

[2] Mostly private tubewells in Pakistan use 10–24 hp diesel engines as the prime mover. These diesel engines are commonly of two types i.e., 12–16 hp Chinese Diesel Pumps locally called as ''Petter Engine'', probably after the famous brand name of a British engine manufacturer of a bygone era and 10–24 hp Slow Speed Diesel Engine locally called as ''Black (Kala) Engine''.

Increasing demand for food to cope with the rising population and favorable marketing conditions and support prices from government has helped in increasing annual cropping intensities to about 150%. Moreover, many canals can no longer convey their official design capacity, due to siltation and erosion of banks. From the scarcity by design and intensified irrigation practices by farmers, over time canal water availability per unit of irrigated land has become even more limited (Sarwar, 2000).

3. Sources of salts in the Indus Basin

Most of the soil salinity in the Indus Basin is inherent, as it was developed during the process of soil formation. The salts are brought in by the rivers and their tributaries. The average annual salt inflow by the Indus River and its tributaries is estimated at 33 Mg. At present only about 27% (9.0 Mg) of the average annual salt flow is washed out of the system while 73% (24.0 Mg) is retained in the Basin, with 13.6 Mg in Punjab Province and 10.4 Mg in Sind Province. Of the salt stored annually in the Basin, only 2.2 Mg are deposited in a series of evaporation ponds located in the desert area outside the irrigated plain in southeast Punjab and the remaining salts (11.4 Mg) accumulate in the soil profiles in the irrigated lands and in the underlying strata and aquifer (Nespak/MMI, 1993).

Due to an overall shortage of good-quality water, the use of poor-quality groundwater as a supplemental source of irrigation has become routine practice. The large scale exploitation of poor groundwater is another substantial source of salt inflow in the Indus basin, thereby aggravating the problems of soil and water salinization. Fig. 1 demonstrates the spatial distribution of groundwater quality in the Indus Basin, which ranges between marginal to hazardous in the major irrigated areas of the Sind Province (Qureshi et al., 2004). An estimated 28.2 Mg of salt are annually brought to the surface by the extensive tubewell pumping taking place in the Basin. Most of the groundwater contribution is in the Punjab (24.7 Mg) with a further 3.5 Mg in the Sind Province. As a result, salt accumulation in Sind province is much less than the Punjab province.

Most of the salts pumped through deep tubewells are fossil salts deposited in the deeper strata and aquifers in the course of the formation of the Indus Plain. Groundwater overdraft has also created serious groundwater quality issues even for shallow tubewells in many areas. Therefore secondary salinization associated with the use of poor quality groundwater for irrigation has further compounded the soil salinity problem. Although the salt accumulation in the Sind province is much less than in the Punjab, it needs more efforts to desalinize, dispose off and reclaim saline lands due to high and poor quality groundwater tables and lack of drainage and disposal facilities. Being close to the sea, several feasible options exist. The problems of drainage water disposal are much more serious in the Punjab. Presently, most of the saline drainage effluent is being reused and then disposed of in the evaporation ponds. However, this is not a feasible long-term solution and in future, more environmentally friendly solutions need to be sorted out.

4. Extent of saline and waterlogged soils in the Indus Basin

The introduction of a large scale irrigation network without adequate drainage altered the hydrological balance of the Indus Basin. At the time of construction of the network, the groundwater table depth in different canal command areas

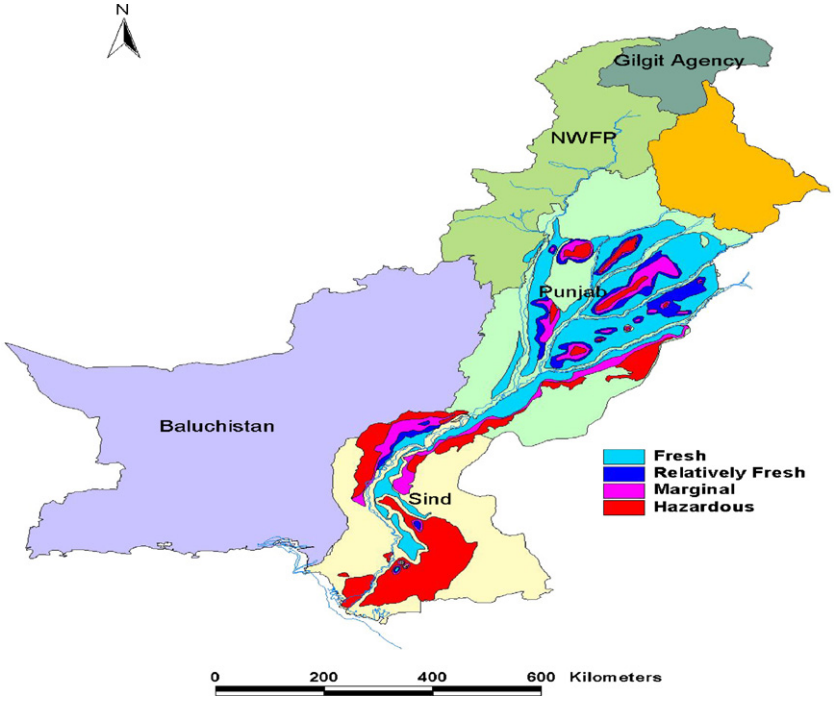


Fig. 1 – Spatial distribution of composite groundwater quality in the Indus Basin of Pakistan (Qureshi et al., 2004).

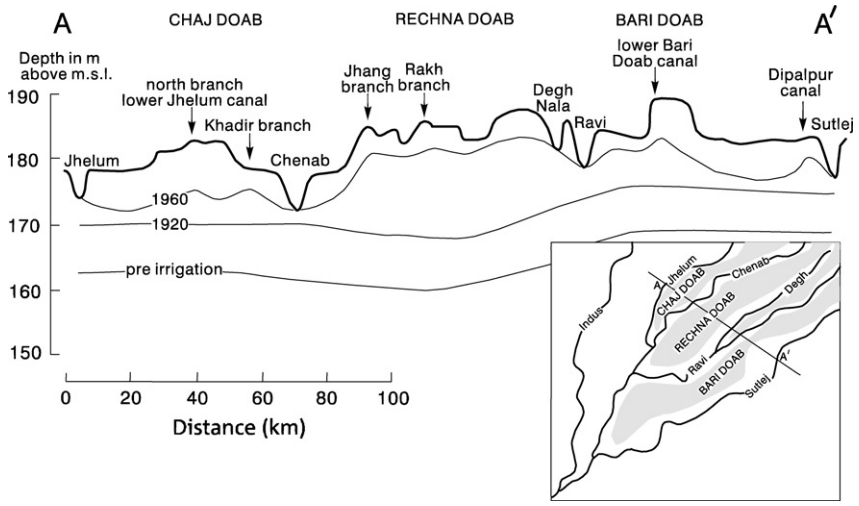


Fig. 2 – The rise in the groundwater level after the introduction of canal irrigation in the Punjab Province of Pakistan. The groundwater profiles are shown for pre-irrigation period, and for the years 1920 and 1960 (Wolters and Bhutta, 1997).

ranged between 20 and 30 m below the soil surface. Therefore the need for provision of drainage for salt management as a part of the irrigation system was not envisaged at that time. The topography of the Indus Basin is relatively flat and characterized by a lack of any well-defined natural surface and subsurface drainage.

The continuous expansion of the irrigation system, blockage of drainage, seepage from unlined earthen canals and percolation from irrigation fields has caused the groundwater table to rise to within 1.5 m of the soil surface in vast irrigated areas. This has created waterlogging and, consequently, soil salinity. These problems are more serious in areas where groundwater is saline. As a typical example, the rise in the groundwater level after the introduction of the irrigation system in the Punjab is shown in Fig. 2 (Wolters and Bhutta, 1997).

The groundwater level in the Indus basin fluctuates seasonally. In general, groundwater levels are deepest at the end of the dry season (May–June) and shallowest immediately after the wet season of monsoon rainfall (September). Estimates show that after the monsoon season, about 4.7 million ha (30% of the irrigated area) have groundwater levels within 1.5 m of the soil surface. Prior to the monsoon period, this area is reduced to about 2 million ha, which is 13% of the irrigated area (Tarar, 1995). The Punjab Province has about 25% of its irrigated area severely waterlogged and Sind has about 60% in the same category. Due to the presence of shallow and saline groundwater, about 40,000 ha are annually abandoned within the Indus Basin due to secondary salinization (WAPDA, 1989). Fig. 3 shows that about 46% of the irrigated land has groundwater levels deeper than 3 m and this proportion is not affected by the season.

According to latest estimates, about 21% of the irrigated land is affected by varying levels and types of salinity. Of this, 7% is severely affected and 4% is moderately affected by salinity. Of course, the scale of the problem of salt accumulation in the root zone is even greater in places where saline groundwater is used for irrigation. Province-wise distribution of salt-affected soils is given in Table 1, which shows that problems of soil salinity are more serious in the lower part of the Indus Basin where about 54% area is saline. The extent of salinization keeps on changing due to the dynamic nature of the problem in the Indus Basin. The main reason for higher soil salinity in the lower part is low rainfall, high evaporation rates, and shallow and saline groundwater.

The problems of land degradation and water quality deterioration in the Indus Basin do not stem from salinity alone. About 70% of the tubewells installed in the basin to supplement surface supplies, are pumping sodic water, which contains high concentrations of Na^+, carbonate (CO_3^{2-}) and bicarbonate (HCO_3^-). Irrigation water with elevated levels of Na^+, CO_3^{2-} and HCO_3^- gradually induces sodicity in the soil system. Sodic soils are an important category of salt-affected soils, exhibiting unique structural problems as a result of certain physical processes (slaking, swelling, and dispersion of clay) and specific conditions (surface crusting and hard setting) (Shainberg and Letey, 1984; Sumner, 1993; Qadir and Schubert, 2002). These problems can affect water and air movement, plant-available water holding capacity, root penetration, seedling emergence, runoff and erosion, as well

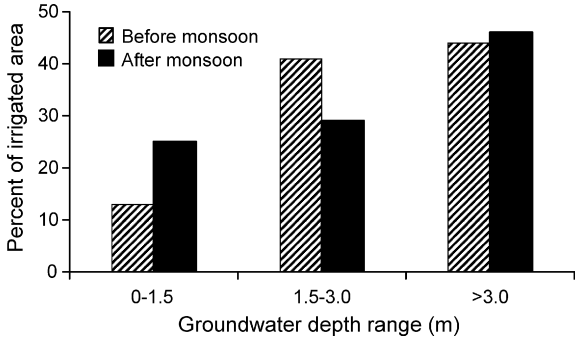


Fig. 3 – Seasonal effects on groundwater depths in the Indus Basin of Pakistan. Seasonal changes and rainfall greatly impact shallow watertable (<1.5 m) areas, but have little impact in areas with deep watertable (>3 m) (Sarwar, 2000).

Table 1 – Province-wise distribution of cultivated area and salt-affected area (WAPDA, 2003)

	Provinces				
	Punjab	Sindh	NWFP	Balochistan	Pakistan
Cultivated area (Mha)	12.27	5.65	2.11	1.84	21.87
Salt-affected area (Mha)	1.234	3.04	0.11	0.12	4.50

as tillage and sowing operations (Oster and Jayawardane, 1998). Salt-affected soils of the Indus Basin are usually classified into four categories (Qureshi and Barrett-Lennard, 1998). The area affected and the characteristics of these four categories are given in Table 2.

The problems of waterlogging and salinity are not just agricultural problems, but they do affect the social fabric of Pakistani society. About 75% of the population and about half of the Gross National Product (GNP) are directly or indirectly related to the agricultural sector. Together waterlogging and salinity rob Pakistani farmers of about 25% of their production potential for major crops, valued at an estimated cost of Pakistan Rupee 14 billion (1 US$ ≈ 58 Pak Rs. in 1999) annually in lost production (Haider et al., 1999). Waterlogging and salinity have adverse social and economical effects on rural communities causing poor living standards and health problems for humans and animals. Loss of agricultural production and limited on-farm income generation activities have forced the local population to migrate to nearby cities to earn their living.

5. Management of salinity and waterlogging through engineering strategies

The threat of waterlogging and soil salinity in the Indus Basin was recognized soon after the introduction of large scale irrigation systems. The first observation wells to monitor the effect of irrigation on the groundwater table depth were installed as early as in 1870. Based on these studies, various remedial measures were tried. These measures included closure of canals in the monsoon season, construction of surface drains in waterlogged areas and lowering of full supply levels of canals (Ahmad and Chaudhry, 1988). As a result, about 15,000 km of surface drains have been installed (Afzal, 1992). However, none of these measures provided more than a local or temporary relief and the regional problem of waterlogging and salinity continued to increase in severity.

5.1. *Vertical drainage projects*

The first detailed survey of groundwater depth and salinity was conducted in 1953–1954 with the collaboration of the US Geological Survey. This survey found that about 4.9 million ha were affected by severe salinity and 4.6 million ha were covered with salinity patches. This survey formed the basis for the Salinity Control and Reclamation Projects (SCARPs) and the decision to go ahead with vertical drainage by initiating a large-scale public tubewell program. As a result, about 14,000 tubewells (covering about 2.6 million ha of irrigated land) with an average capacity of 80 l s^{-1} were installed in fresh groundwater areas to control waterlogging and salinity problems through lowering of the groundwater table. The pumped groundwater was discharged into the existing canal system to increase irrigation supplies at the farm gate. The total coverage of the SCARP program was 8 million ha with an estimated cost of US$ 2 billion (Badruddin et al., 1999).

The demonstration of SCARP tubewells led to a proliferation of private tubewells in the 1970s and 1980s with the capacity of about 28 l s^{-1}. SCARP tubewells managed to lower groundwater level below 1.5 m in an area of 2 million ha and below 3 m in 4 million ha, thereby alleviating the problem of waterlogging significantly. It also reduced the salt-affected soils from 7 million ha to 4.5 million ha (Table 1). The additional water supplies made available by the SCARP tubewells and the associated lowering of groundwater table resulted in increased cropping intensities in SCARP areas. The cropping intensities were increased from 80 to 120% in most of

Table 2 – Extent, classification, and characteristics of salt-affected soils in Pakistan (Based on Qureshi and Barrett-Lennard, 1998)

Soil type	Extent (million ha)	Characteristics
Slightly saline-sodic	0.7	Slight salinity-sodicity problem, occurring as patches (about 20% of the area) in cultivated fields. Thus, about 3.5 million ha (0.7/0.2 = 3.5 million ha) of agricultural land is affected by this problem.
Porous saline-sodic	1.9	Saline-sodic throughout the root zone, but are porous and pervious to water; loamy to clayey in texture; respond to both chemical and biological amelioration efforts.
Severely saline-sodic	1.1	Severely saline-sodic with high groundwater levels; dense and nearly impervious to water.
Soils with sodic tubewell water	2.3	Sodic soils developed from irrigation with sodic tubewell water containing high concentrations of carbonates (CO_3^{2-}) and/or bicarbonates (HCO_3^-); about 70% of the tubewells pump sodic and/or saline-sodic water in the Indus Basin

the SCARP areas (IWASRI, 1998). Although poverty considerations were not given a high priority in the SCARP projects, these projects helped in alleviating poverty through reduced waterlogging and salinity and enhanced cropping productivity (Qureshi et al., 2003).

Despite these successes, the operation of SCARP tubewells became a huge burden for the government due to high operational and maintenance costs. Poor maintenance over time resulted in inefficiency and in some cases completely closed down the operation of SCARP tubewells. In addition, the uncontrolled use of groundwater for irrigation in some areas without commensurate adjustment to the SCARP extraction rates led to excessive lowering of groundwater level and intrusion of saline groundwater into freshwater aquifers. This not only deteriorated the quality of groundwater but also increased the pumping cost, some of which would have been unnecessary. The normal soils also started showing initial symptoms of salinization due to the degraded quality of the groundwater. As a result, farmers demanded that SCARP tubewells in the affected areas and, as a result, operation of SCARP tubewells was restricted in many areas.

5.2. *Horizontal drainage systems*

In the middle of 1970s, it was realized that circulating saline water through vertical drainage only serves to aggravate the problem. Therefore, thinking shifted towards horizontal (pipe) drainage systems particularly in saline groundwater areas. Under Pakistani conditions, the capital costs of horizontal drainage systems are some 10 times more expensive than tubewell systems (roughly US$ 1000 ha^{-1} compared to US$ 100 ha^{-1}) (Bhatti, 1987). Despite these higher costs, the main reasons to introduce horizontal drains in saline groundwater areas were based on the assumptions that the drainage water volume will be reduced and the quality will improve. Improved drainage water quality would reduce disposal problems and increase the possibility of using drainage water for irrigation.

In Pakistan, subsurface drainage for waterlogging and salinity control was first introduced in the East-Khairpur project. The project was constructed between 1977 and 1985. After this, pipe drains were installed in the Mardan SCARP project and the Fourth Drainage Project (FDP). At the time of construction of these projects, there was limited experience with the planning and design of subsurface drainage systems for the arid and semi-arid conditions of Pakistan. Therefore these projects were designed by applying the drainage design criteria as used for humid areas. These criteria are mostly based on the steady-state equations of Hooghoudt and Ernst (see Ritzema, 1994), which assume steady-state moisture and solute fluxes occurring in the unsaturated zone i.e. the design of drain depth and drain spacing for *a priori* chosen drain discharge rate and permissible depth to groundwater table.

The drain discharge rate for the above-mentioned drainage projects was calculated using simple steady-state water balance approach. For the East-Khairpur drainage project, a drain discharge rate of 3.5 mm d^{-1} was calculated (van Someren and Th. Boers, 1978). For the Mardan SCARP project, the design drainage rate was estimated to be 3.2 mm d^{-1} and for the Fourth Drainage Project (FDP), a value of 2.44 mm d^{-1} was utilized. These estimated drain discharge rates were then used to calculate the required lateral spacing. The average drain depths of these three projects range from 1.95 m for East-Khairpur to 2.44 m for the FDP and were mainly selected on the basis of minimum costs (Smedema and Aart, 1992). The comparative studies of the above three completed projects have shown that all three designs are rather conservative with high drainage intensities. Discussions with farmers at Mardan and FDP projects confirmed that in the areas already drained the demand for surface irrigation has increased (Vlotman et al., 1990). Farmers in these areas blocked the drainage pipes to control excessive drainage and reduce the demand for surface irrigation. For all these projects, drainage water from collectors is discharged into a sump and from there water is pumped into the nearby surface drains. These horizontal drainage projects also require funds for operation and maintenance, which was again a burden for irrigation department as farmers were not willing to share the cost.

Smedema (1990) proposed a drainage design discharge of about 2.0 mm d^{-1} for (semi-) arid conditions. But this criterion was not considered suitable for local Pakistani conditions as the situation varied from one location to the next. Therefore it was realized that there is a need to refine Smedema's findings for local conditions. Since the completion of the East-Khairpur drainage project, nine more pipe drainage projects have been designed, of which several have been completed and others are in various stages of completion. Since the design of the first drainage project, the value of design drain discharge rate has gradually reduced from 3.5 mm d^{-1} at East-Khairpur to 1.3 mm d^{-1} at Khushab Project (Fig. 4). Although local conditions such as annual rainfall or seepage loss from canals influence the drain discharge rates, the general trend towards lower drain discharge rates can be explained by the experience designers have built up over the years who now feel more confident to design for lower drain discharge rates (Boers and Zuberi, 1995). This demonstrates the importance of the 25-years of experience in the field of drainage that Pakistan has now amassed and underscores the need to ensure that the skills and knowledge be retained and enhanced to address the waterlogging and salinity challenges in the Indus basin.

One of the major bottlenecks in the successful operation of drainage projects for salt management has been the safe disposal of saline drainage effluent. Only a small amount of

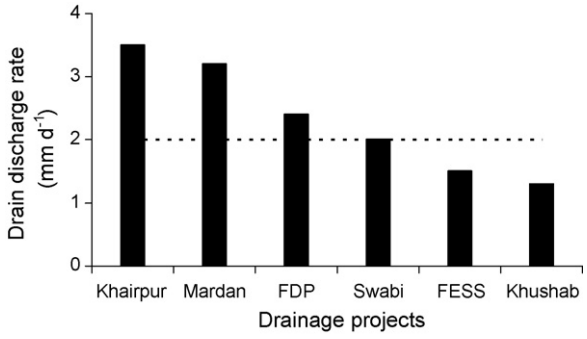


Fig. 4 – Drain discharge rates used for different drainage projects in Pakistan (derived from data of Bhutta et al., 1994). The dotted line represents the value proposed by Smedema (1990) for semi-arid regions. FESS represents Fordwah Eastern Sadiqia South drainage project.

saline effluent generated by drainage systems is re-used for irrigation to grow salt tolerant crops and the rest is discharged into surface drains or adjacent lowland areas. In 1980s, Pakistan launched a mega drainage project to alleviate waterlogging and salinity from left side of the Indus River by discharging highly saline drainage brine to the Arabian Sea. The project was named as Left Bank Outfall Drain (LBOD). LBOD was completed in 1995 with a total cost of about 636 million US dollar. Out of the total cost, 60% was provided by a consortium of international donors and the rest 40% comes from government of Pakistan. This project involves the construction of 1950 km long surface drains, 2000 tubewells, 5000 structures and 2000 km long buried pipes and improvement in irrigation supplies. LBOD provided drainage to 516,000 ha saline area (1230 US$ per ha).

LBOD project showed remarkable results in lowering groundwater tables, increasing cropping intensities and crop yields. Many farmers started returning to their land to grow crops as water tables were lowered through subsurface tiles and tubewells and emptied into a drain (World Bank, 1997). However, before its full-scale operation, this project was criticized for not being environmentally friendly, primarily because of degradation of lands adjacent to the drains due to heavy seepage and damages to aquatic life. Moreover, operational costs of the drains presented yet another reoccurring financial burden on the government, jeopardizing the long-term sustainability of this project. In order to address environmental issues, another major drainage project was initiated in 2001 to provide drainage to the saline areas located on the right side of the Indus River. This project was called as Right Bank Outfall Drain (RBOD). The estimated cost of this project is 250 million US dollars with a completion time of 4 years. However, completion of this project has been delayed due to administrative and financial constraints. It is hoped that the completion of this problem will solve the drainage and salinity problems in Sind province.

Once these projects were envisaged, Sind province objected that the drainage water from irrigated lands of Punjab should not pass through Sind on its way to Sea because it would aggravate the existing salinity problems. As a consequence, Punjab has had to redirect its drainage discharge into the natural sand depressions located in south of Punjab. The expectation was that these depressions serve as natural evaporation ponds and the disposed water be evaporated due to extremely high temperatures. However, this did not happen as the discharge was much greater than the evaporation rates. The situation became worse when seepage from these ponds started affecting adjacent agricultural lands, and it was then proposed that the ponds be linked to the LBOD and to the sea. However, this was the original point of dispute between the two provinces and has yet to be resolved.

6. Future perspectives: Managing the problem—building livelihoods

Despite significant investments, waterlogging and salinity remain as major threats to the livelihoods of millions of poor in Pakistan, and to national food security of the country as a whole which has to produce more food for a growing population. On top of this, the Indus basin is a closing basin, which means that despite the impression that there is too much water in the basin most of it is already being utilized and competition from other sectors is likely to increase. Over the course of the last three decades, the public and private sectors have undertaken major activities to meet the challenges yet success has remained elusive. A major reason that these constraints remaining intractable is that given the scale of irrigation development within the Indus it is imperative that salinity control and drainage be considered within the context of the basin and the overall salt balance as the discharge from one location affects the next location downstream at all scales, most notably inter-provincial.

Within the context of the overall basin and the overall salt balance, the salt management issues are very complex and no single straight forward solution applies. To sustain and improve the productivity of irrigated agriculture, the integration of irrigation and drainage is absolutely necessary because irrigation management and drainage problems are closely interlinked through: (i) over or inefficient irrigation as a cause of waterlogging and (ii) the relationship between irrigation management and effluent disposal. Minimizing drainage effluent by improving irrigation efficiency and re-use of drainage water at the farm level while maintaining the health of the soil is an attractive alternative. Where relevant, farmers need to have access to the knowledge and inputs for reclaiming salt affected lands through physical, chemical, and biological approaches. Timely availability of farm inputs such as good-quality water, salt-tolerant germplasm, and introduction of groundwater abstraction regulations and promotion of saline agriculture through crop diversification options such as salt-tolerant medicinal and aromatic plant species can improve the capacity of individual farmers to be productive.

The institutional arrangements regarding drainage and irrigation management need to be developed to remove the constraints at the farm, system, provincial and basin scales, and to ensure that the various interventions and components of the system are integrated as much as practicable. Presently farmers pay for some of the operation and maintenance costs of the surface irrigation costs but have so far been unwilling to pay for the drainage systems as they view it as a government responsibility. One of the major reasons for this behavior of farmers is that the drainage cots are considerably higher, because unlike canals, surface drains are not self maintaining and sub surface drainage involves huge pumping costs. The Operation, Maintenance and Recovery (OMR) cost of the LBOD Drainage system in Sind has been estimated to be US$ 43/ha. This amount is too big for farmers to pay.

As discussed above, the financial resources from the Government is not always reliable. While in the short term increased investments from the public sector will be necessary to ensure that the systems are fully operational, such investments need to be made with the medium term view of securing the revenue stream to avoid deterioration again. In order to encourage participation of farmers in sharing the costs of operation and maintenance of irrigation and drainage systems, government introduced Water User Associations (WUAs) act in 1981–1982 and Provincial Irrigation and Drainage Authorities (PIDAs) act in 1997. Under the PIDAs, the operation and maintenance of individual canal system is

to be entrusted to autonomous self-accounting Areas Water Boards (AWBs), which would entrust the O&M of the distributaries, minor and canals to Farmers Organizations (FOs). Under these arrangements, many Farmers Organizations (FOs) and Water Users' Associations (WUAs) have also been formed and some are functioning. The process of implementing the institutional reforms has slowed the participation of farmers in the management of O&M activities because integration of existing laws into a single comprehensive law for future arrangements has not yet been finalized.

The left and right bank outfall drains (LBOD and RBOD) can play a vital role in reclaiming saline soils and providing an outfall for evacuated salts from existing and future drainage projects. However, for successful operation of these drains, better coordination among different federal and provincial government agencies to resolve inter-provincial water allocation and related issues would be of paramount importance. In the absence of such arrangements, capital costs of installing these systems will go wasted and salt management and consequently sustainability of irrigated agriculture in the Indus Basin will remain a challenge.

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